Mixing method: Specific Heat of a solid

① Aim of Experiment :-

Determine the specific heat (sf) of an unknown solid using the mixing metod

② formula & Terms

$$Sf = (mw Sw + mc Sc) \cdot \frac{Tf+w - Tw}{mf\ (Tf - Tf+w)}$$

Term	Description	Units
mf mw mc	Mass of fragments, water, Calorimeter	gm
sf Sw Sc	Specific Heat, (fragments, water, cal.)	cal/gm.°C
Tw	Initial Temperature of water	°C
Tf	Initial Temperature of Heated fragments	°C (approx 90°C)
Tf+w	final Equilibrium Temperature	°C

③ Experimental Data Table

Mc = —	gm
Mf = —	gm
Mw = —	gm
Sw = —	cal/gm.°C
Sc = —	cal/gm.°C
Tw = —	°C
Tf = —	90 °C
Tw+f = —	°C

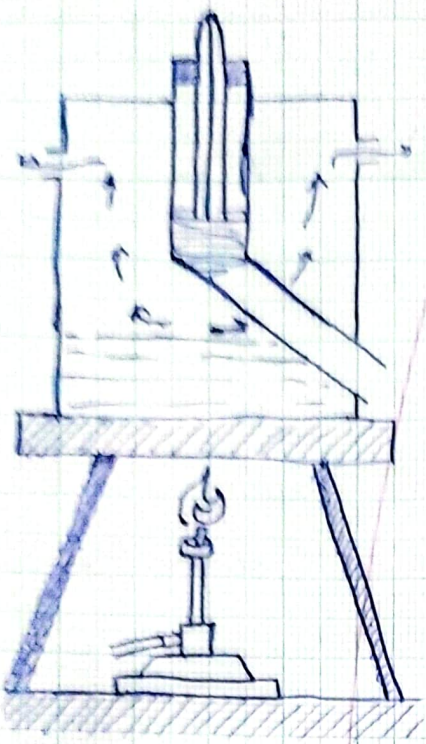
④ final Calculation

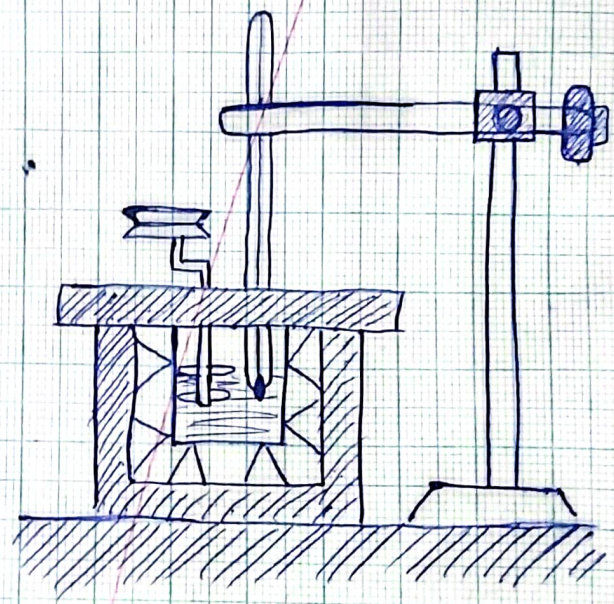
sf = _____ Cal/gm.°C

$$Sf = \frac{(mw Sw + mc Sc)(Tf+w - Tw)}{mf\ (Tf - Tf+w)}$$

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① Experimental Apparatus Setup





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